

How to Read/Summarize/Write a Research Paper

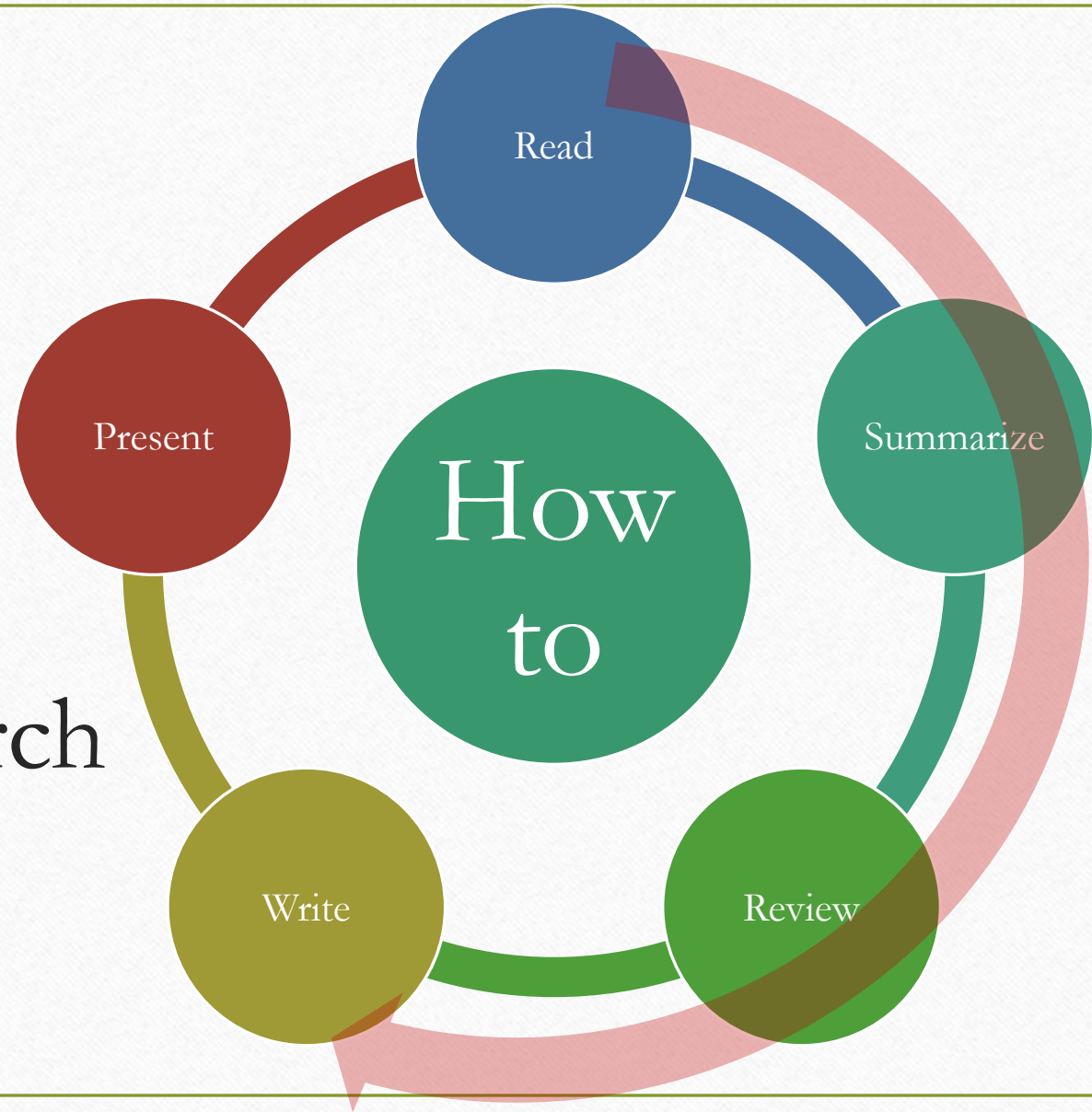
CSE 300: Technical Writing & Presentation

Course Teachers: MSB, MTB, RR, MSRJ, AW

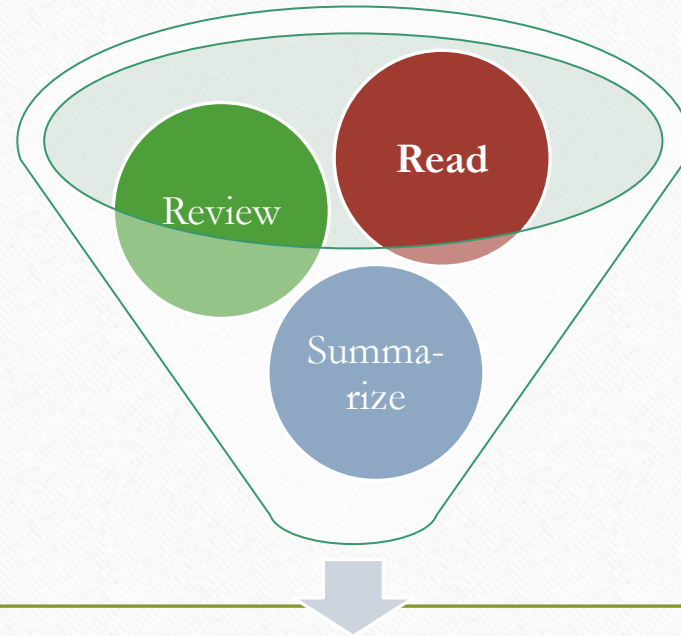
Presenter: MSRJ



How to ___ a Research
Paper?



Section 1: READ



How to Read a Research Paper?



WHY





Topic

- Right paper?
- Exp. vs. Theor.?

Venue

- Reputed venue?
- Jour. vs. Conf.?

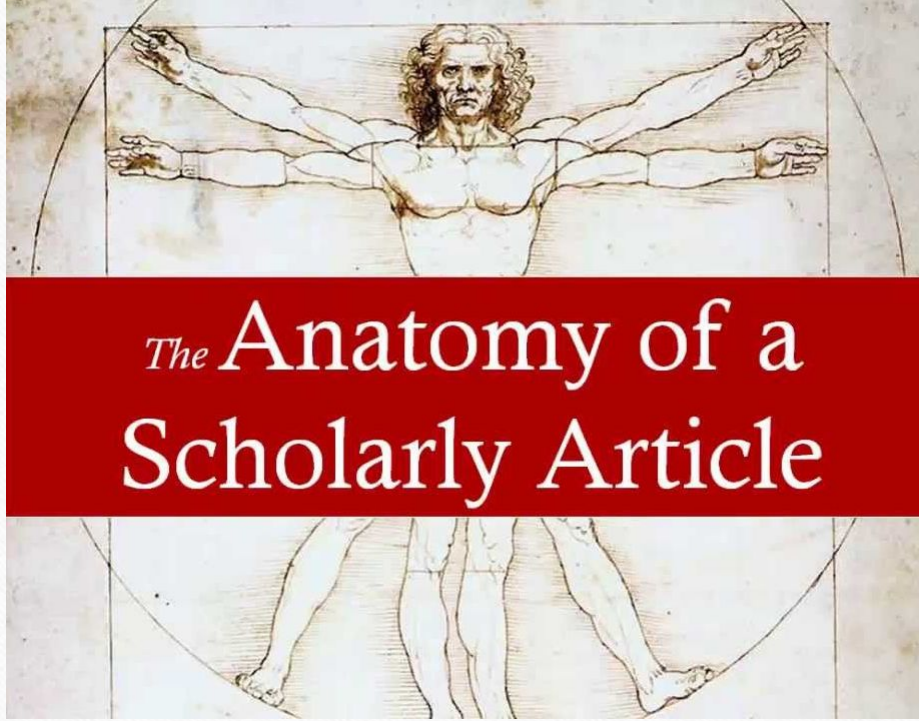
Quality

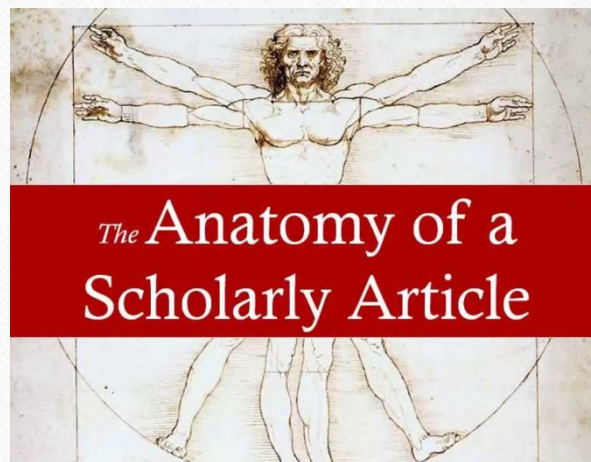
- Citation?
- Impact?

Authors

- Reputed?
- Recent Works?







ANATOMY OF A SCIENTIFIC ARTICLE

ABSTRACT

A very short, clear, and concise summation of the entire paper.

INTRODUCTION

The introduction reveals the purpose of the paper, starting broadly and narrowing down to a specific thesis or research question.

MATERIALS & METHODS

Can be divided into two separate sections. Clearly outlines how the study/experiment was carried out.

RESULTS

Objectively presents key results without interpretation in an orderly, logical, sequence. Includes text, graphs, statistics, etc.

DISCUSSION

Interprets results in light of what was already known about the subject, and to explain new understanding of the problem after taking found results under consideration.

CONCLUSION

Reinforces major claims or interpretations. The author will try to indicate the significance of the major claim/interpretation beyond the scope of the paper.

IMRaD



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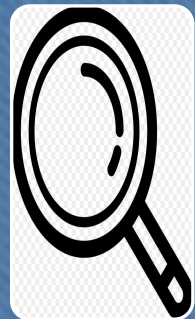
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Multi-Pass Approach

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General
Idea

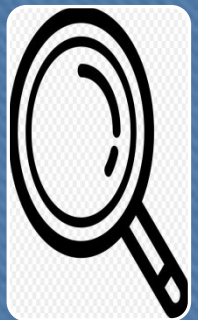
• Pass 1

Content
(No Details)

• Pass 2

Depth

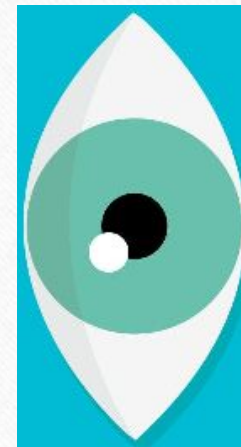
• Pass 3



Multi-Pass
Approach



Pass 1



Bibliography

Abrahams, P.A., K.A. Hargreaves, and K. Berry (2004). *T_EX for the Impatient*. Addison-Wesley, Inc. <http://tug.org/tex/impatient>.
Allen, Robert (2001). *Punctuation*. Oxford University Press. ISBN: 0-19-860490-4.
American Mathematical Society (24 Feb. 2004). *User's Guide for the ansic Package*. Version 2.0.
——— (10 Feb. 2004). *Using the ansic Package*. Version 2.0.
Arnold, Doug (24 Jan. 2004). *The T_EX for Linguists Home Page*. URL: <http://www.essex.ac.uk/linguistics/external/clint/lateXling/>.
Arseneau, Donald (20th Jan. 2004). URL: <http://www.essex.ac.uk/linguistics/external/clint/lateXling/>. Edited as a T_EX document by Robin Fairclough.
Isaksson, Håkan (1998). "Ten T_EX Tricks for the Mathematician". In: *TUGboat 14*. A modern version of this paper is available from

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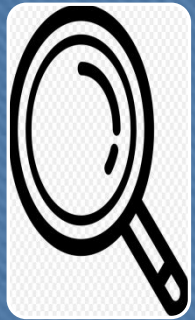
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CONCLUSION

Reinforces major findings. The author will try to indicate a major claim/interpretation of the paper.



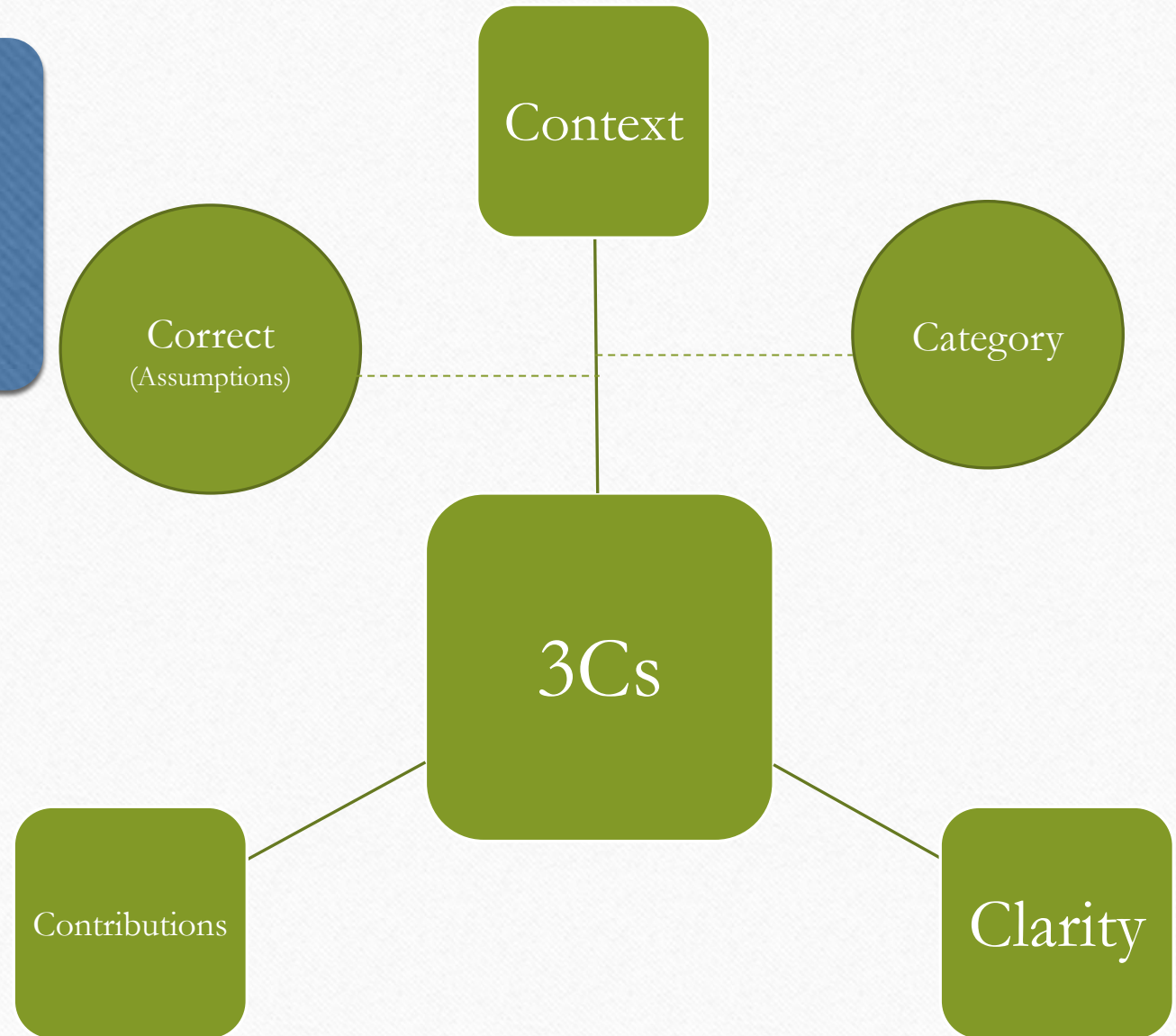
Headings
Sub-heading



Pass 1



CONTINUE



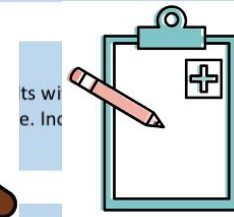


ABSTRACT

INTRODUCTION

MATERIALS & METHODS

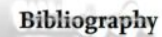
RES

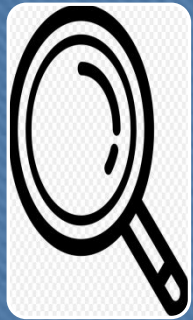


DISCU

CONCLUSION

Reinforces major claims or interpretations. The author will try to indicate the significance of the major claim/interpretation beyond the scope of the paper.





Pass 2



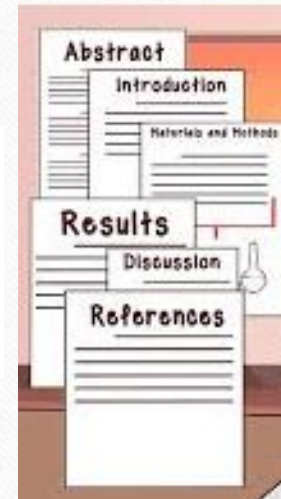
Original Text



Text Summarization



Summarized output



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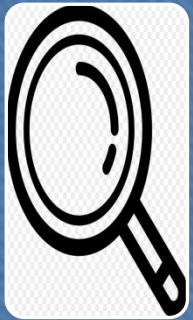
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Set the paper aside

Return to the paper later

Reading background material

Persevere and go to next pass

New

Poor

Jargons

Forward ref.

Not
Understandable

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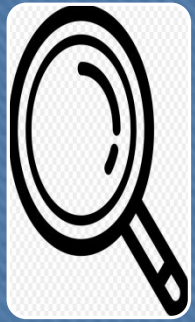
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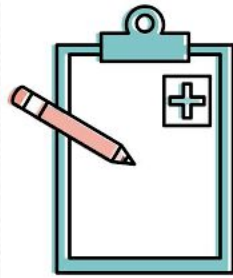
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Pass 3



Make author assumptions

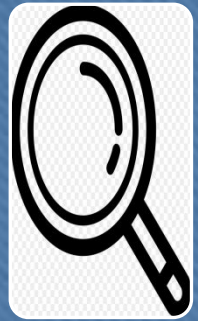
Re-create the work

Identify failings and assumptions

Insight

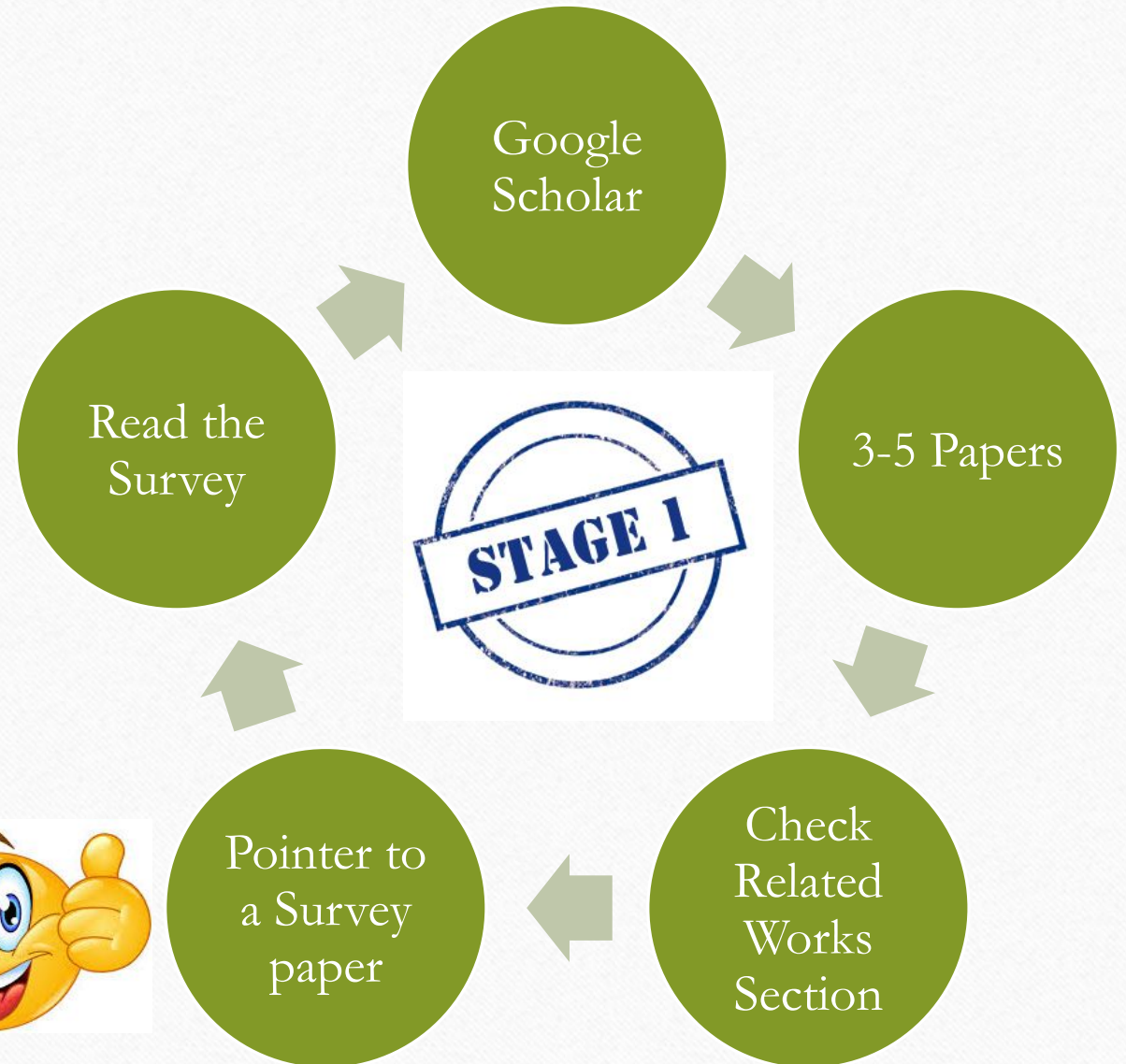
Pinpoint issues

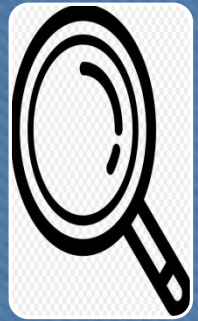
Literature Survey



Pass 3

Literature Survey





Pass 3

Literature
Survey

Top venues

- Check citations
- Repeated Authors
- Find web pages

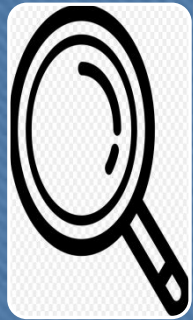
Key Persons

- Where the key persons publish
 - Conferences
 - Journals

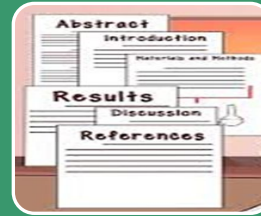
- Download papers
- Read
 - Multi-pass

Proceedings/ Archives





Pass 3



Reconstruct the entire structure of the paper from memory.



Identify its strong and weak points.



pinpoint implicit assumptions, missing citations.



Identify potential issues with experimental or analytical techniques.



Read Critically



Read Creatively



Take Notes



Read
Critically





Read
Creatively



Any Good Ideas?



Idea Extensions?



Influential results?



Possible Improvements?



Take
Notes



Your will Forget

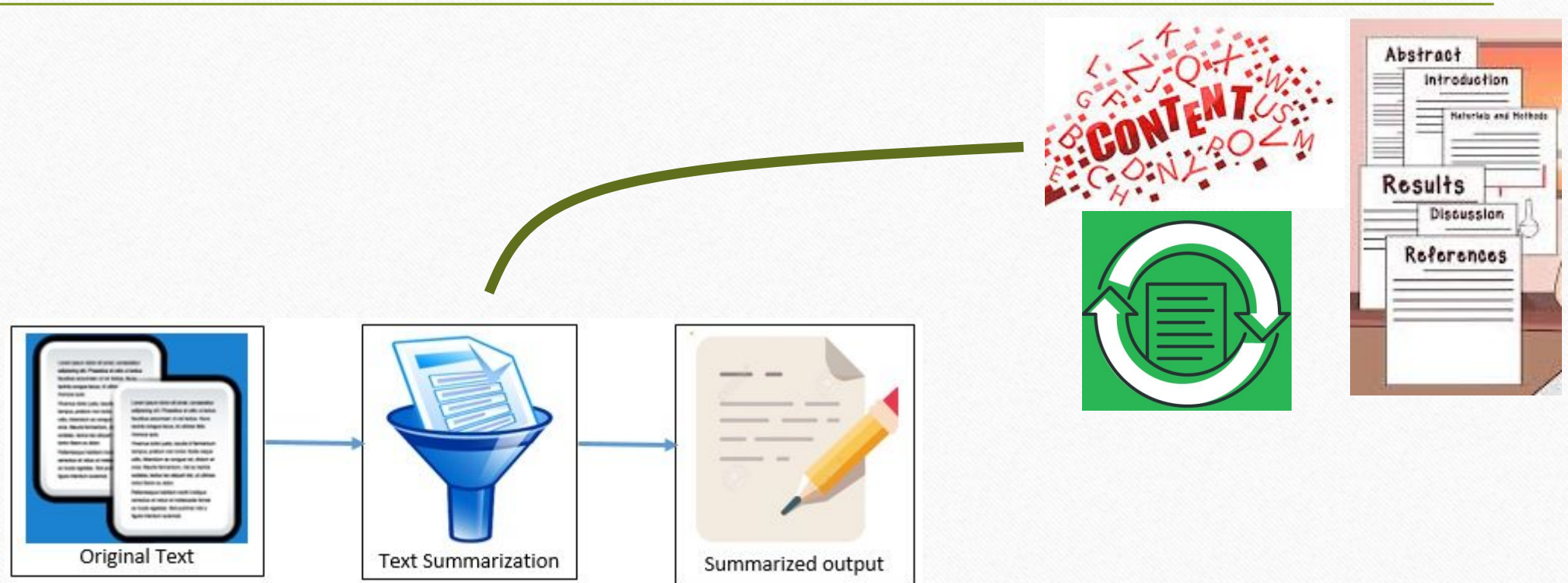


Electronic Notes

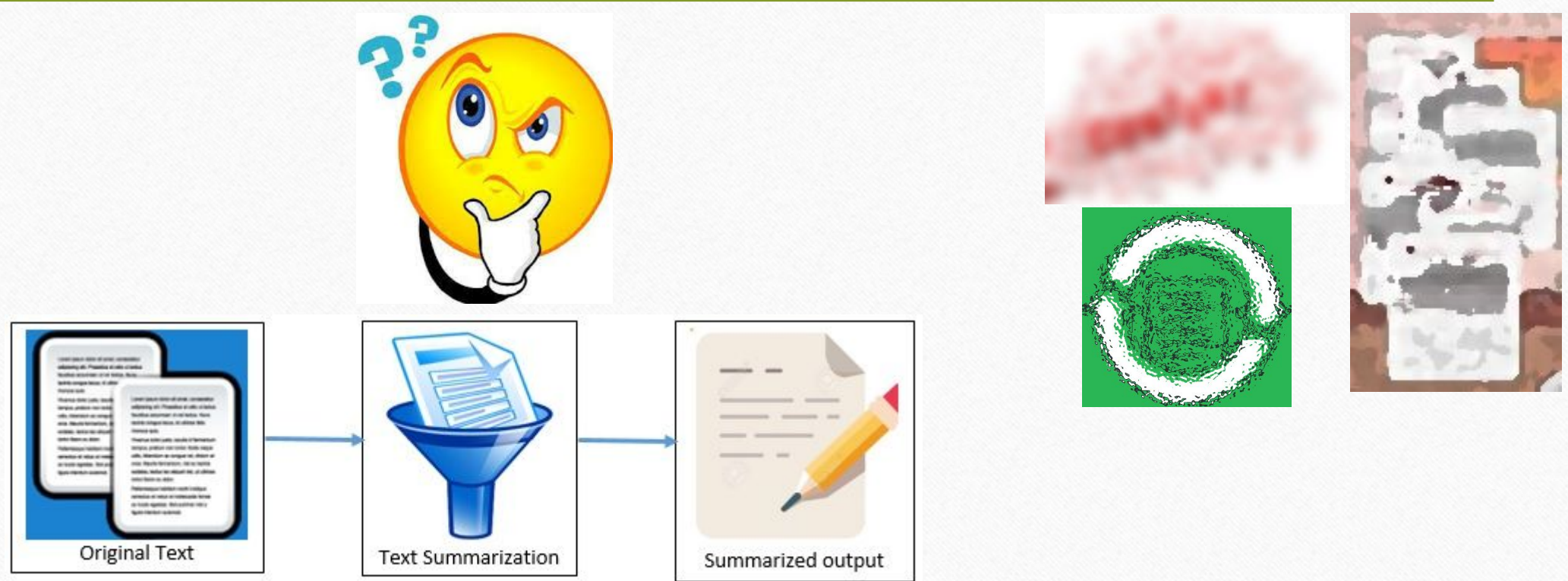


Written Notes

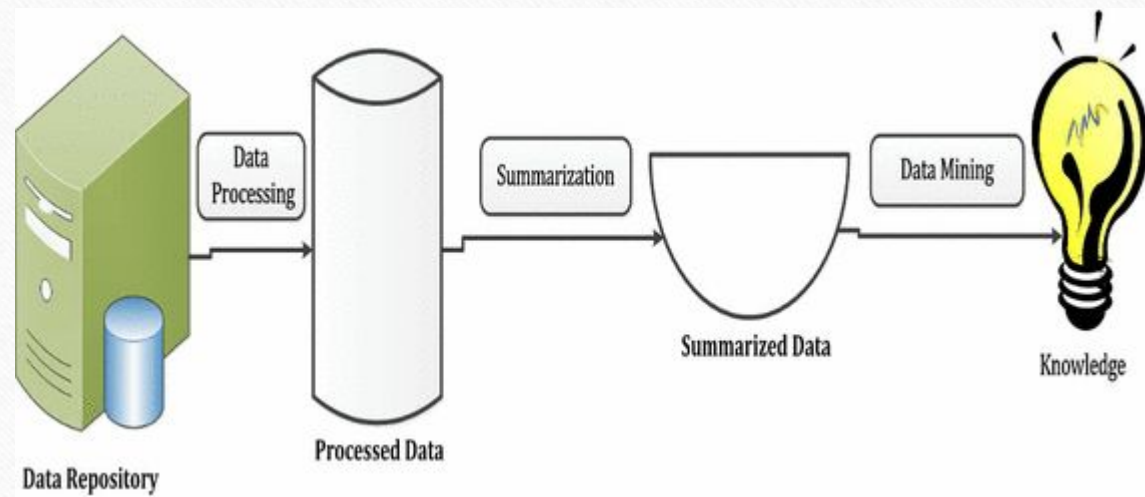
Summarization



Summarization



Automatic Summarization



Section 2: WRITE

How to write a Research Paper?



WHAT



WORTH



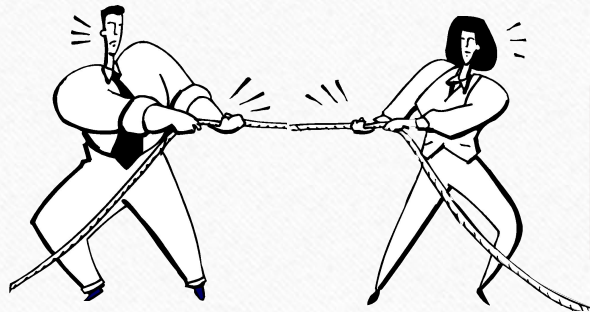
HOW





What do I have to say in the paper?

Is it one or more papers?



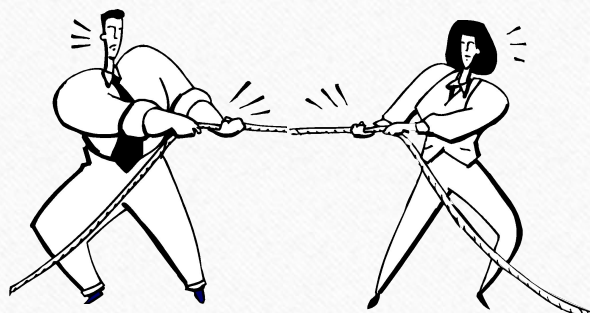
Salami Science???



Putting too much in one paper makes it diffuse and less compelling

Try stating the question and its answer in one simple sentence.

A single question clearly stated with adequate evidence for the answer.



Salami Science

Reporting the results
of a single study in 2 or
more manuscripts.



Wastes Journal space

Time waste for Reviewers

Time waste for Readers



3 Reasons

Publish or
Perish

Quantity;
Not Quality

More
Support

When
Justified?

Different
audience targets

Too long a
manuscript



Is the
paper
worth
writing?

What is in the literature?

It is a lot of work!!!

20-30 drafts!!!

Don't do this unless it is
worth.



OK, So you
want to/need
to write a
paper...

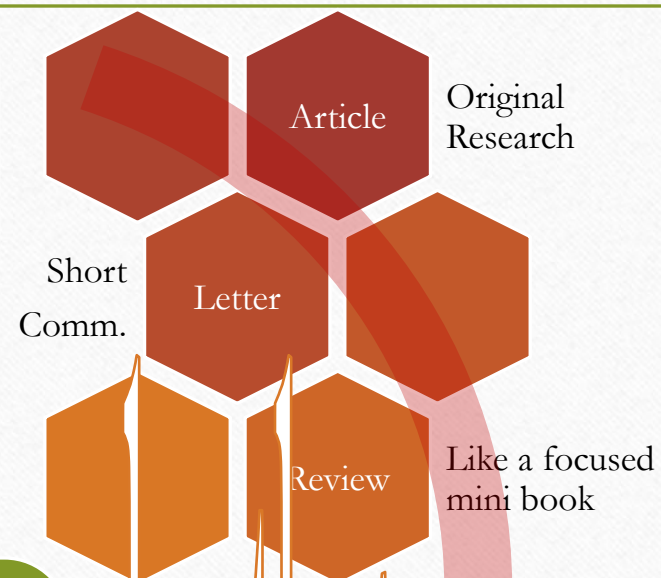
So, How to
Proceed?





More Questions

Format



Journal

Conference

Which J/C

Venue

RESEARCH

Who cares?

Audience



Venue
Scope?

Allowed
Format?

Audience
Match?

Impact?

Take advice

Mentor

senior
researchers

Consequence
of Wrong
Decision

time lost

failure to
publish



Choice
Parameters



Impact
Factor

SCImago
Rank

Editorial
Board

<https://beallslist.net/>



Indexing

...



Research article as a critical Argument

Research Article



Problem Statement
posing a question

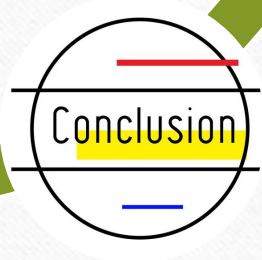


Presentation of
evidence



Assessment of the
validity of the
evidence in the face
of ..

- strengths/weaknesses
- other evidence

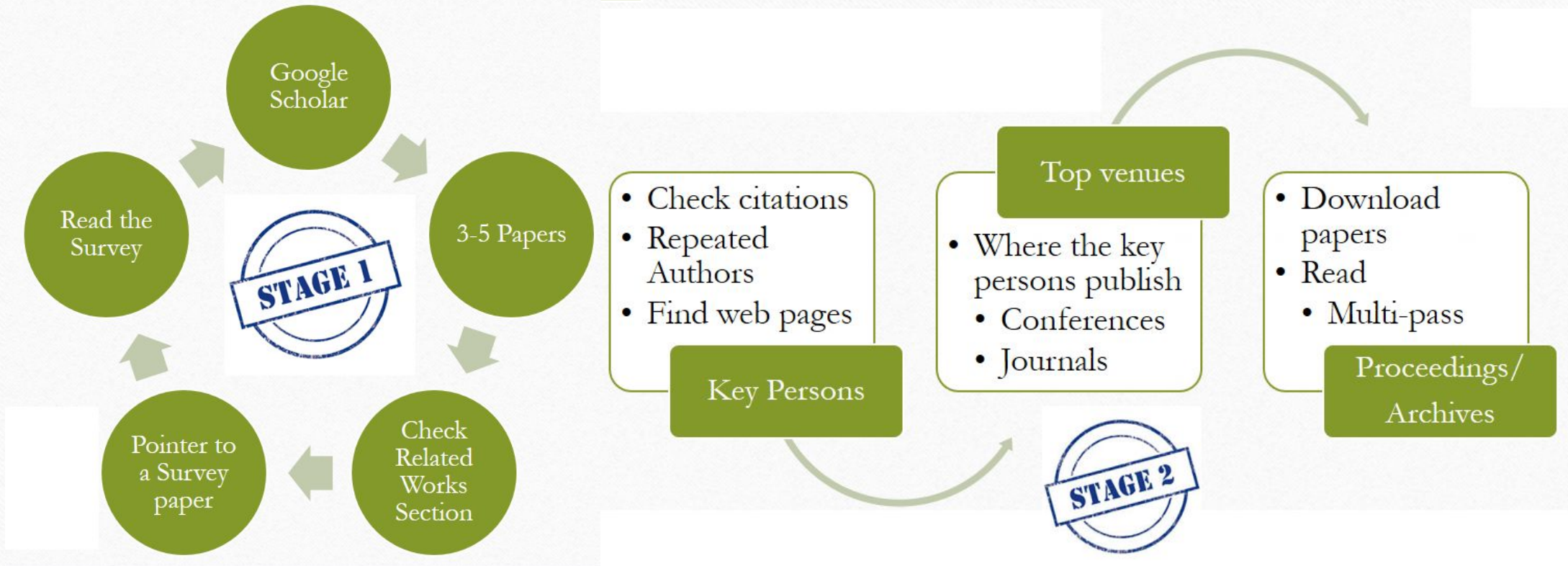


Conclusions



Literature Survey

Recall...





Literature Survey

What has been done and what can you say that's new?

Be thorough in your search:---a high sensitivity/low specificity search.

- Low False -ve
- Misses only a few

High Sens.



- High False +ve
- Too eager to find +ve

Low Spec.





Title Page

Fix earlier than later

Order Alphabetic

Corr.

Senior

First

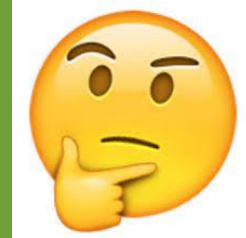
Avoid hurt feeling
Assign
Responsibilities



Title

Tentative

Author
Order



Sources
of
Support

Acknowledgement



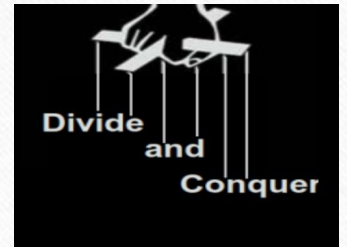
HOW



Create an
outline first

Plan on
multiple drafts

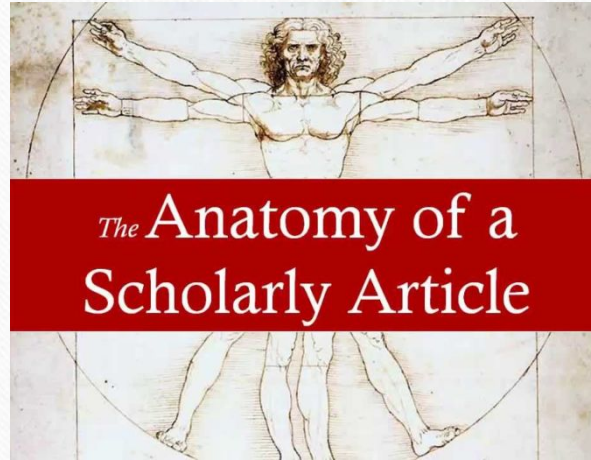
- Filename with dates
- One filename written over with new draft



Tables/figures
early

- prompt more analysis

Deadlines for
you and
coauthors



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Introduction

5Qs

What is the problem

(CS)

Why is it interesting and important?

Why is it hard? (E.g., why do naive approaches fail?)

Why hasn't it been solved before? (Or, what's wrong with previous proposed solutions?)

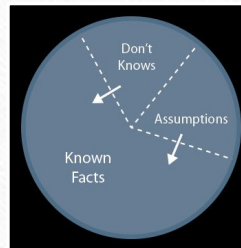
How does mine differ?)

What are the key components of my approach and results?



Draw audience in;
be provocative

Target journal
specific audience



Identify gaps in
knowledge



End with
question/hypothesis





Methods

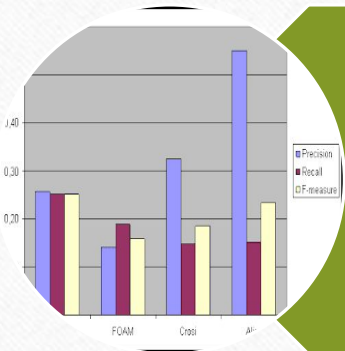
Should include subheadings

Should be comprehensible by
nontechnical reader

Write clearly
enough

Should be replicable

Past tense



Results

Fig. vs.
Tab.

Mix-up



Suppl.



Organize around tables/figures

Past tense

No interpretation; just the facts!

Tables/Figures should be
self-contained



Discussions

1st part
Answer main question/
hypothesis
Interpret the main results
Can take help from the Results section content

2nd, 3rd, ...
Evidence pros & cons
Implications of findings (be conservative)
Other findings of your study
Limitations of your study

Last part
Concluding remarks
Future directions

Could constitute a separate section



Writing Tips

Tense



When quoting *previously published work*, refer to it in *present tense*.

When describing your *own study*, refer to work in *past tense*

Methods and Results in Past tense



Writing Tips

Voice Some branches dictate passive voice only



Some allow (sometimes prefer) active voice

Best practice: Mixed voice



Writing Tips

Brevity Be brief. Make every word count.

If a word does not support your point, cut it out.

Use shorter and more direct phrases wherever possible.



Writing Tips

Section

Paragraph

Sentence

Word

Ask what
role it serves

If not, delete
it

For each
section
in the paper

Whether it
contributes
to the paper's
main point



Writing Tips

Section

Paragraph

Sentence

Word

Does it have a single point

If not, rewrite or delete it.

For each paragraph in a section

Whether it contributes to the goal of the section.



Writing Tips

Section

Paragraph

Sentence

Word

Does it
have a
single point

If not,
rewrite or
delete it

For each
sentence
in a para

Whether it
strengthens
the para



Writing Tips

Section

Paragraph

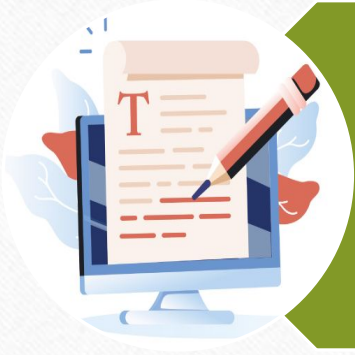
Sentence

Word

Whether it
strengthens
the point

For each
word in a
sentence

If not,
rewrite or
delete it



Proofread

Are terms used consistently throughout?

Do numbers in abstract match numbers in text and tables?

Do citations in text match references?

Are Syntax and Grammar acceptable



Abstract

Like a
paper

- requires many drafts
- most oriented to presenting argument concisely

Mini-
paper

Introduction (usually 1-2 sentences)

Methods (often longest part)

Results (no tables/figures)

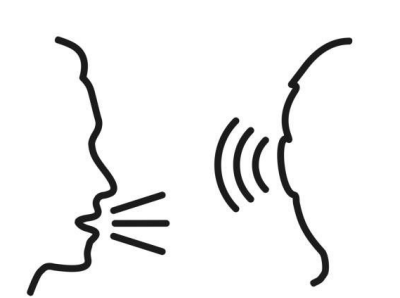
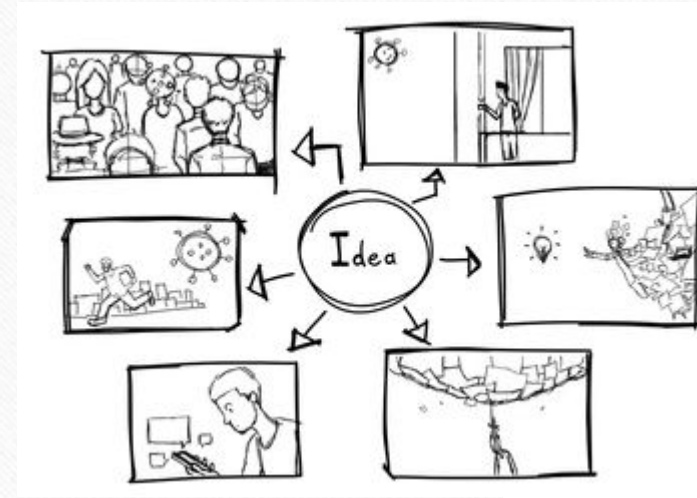
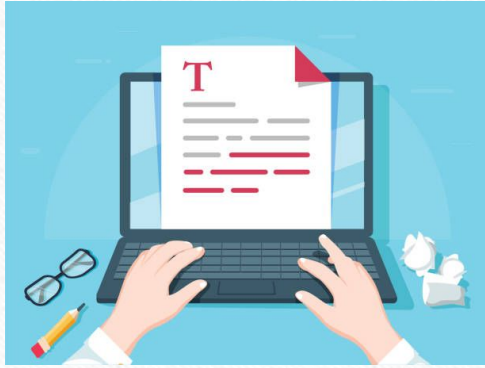
Discussion is limited to concluding statement

Section 3: Seven Suggestions

Writing a CS Paper

Seven simple, actionable suggestions that will make your papers better.

Courtesy: Simon Peyton Jones
Microsoft Research Cambridge



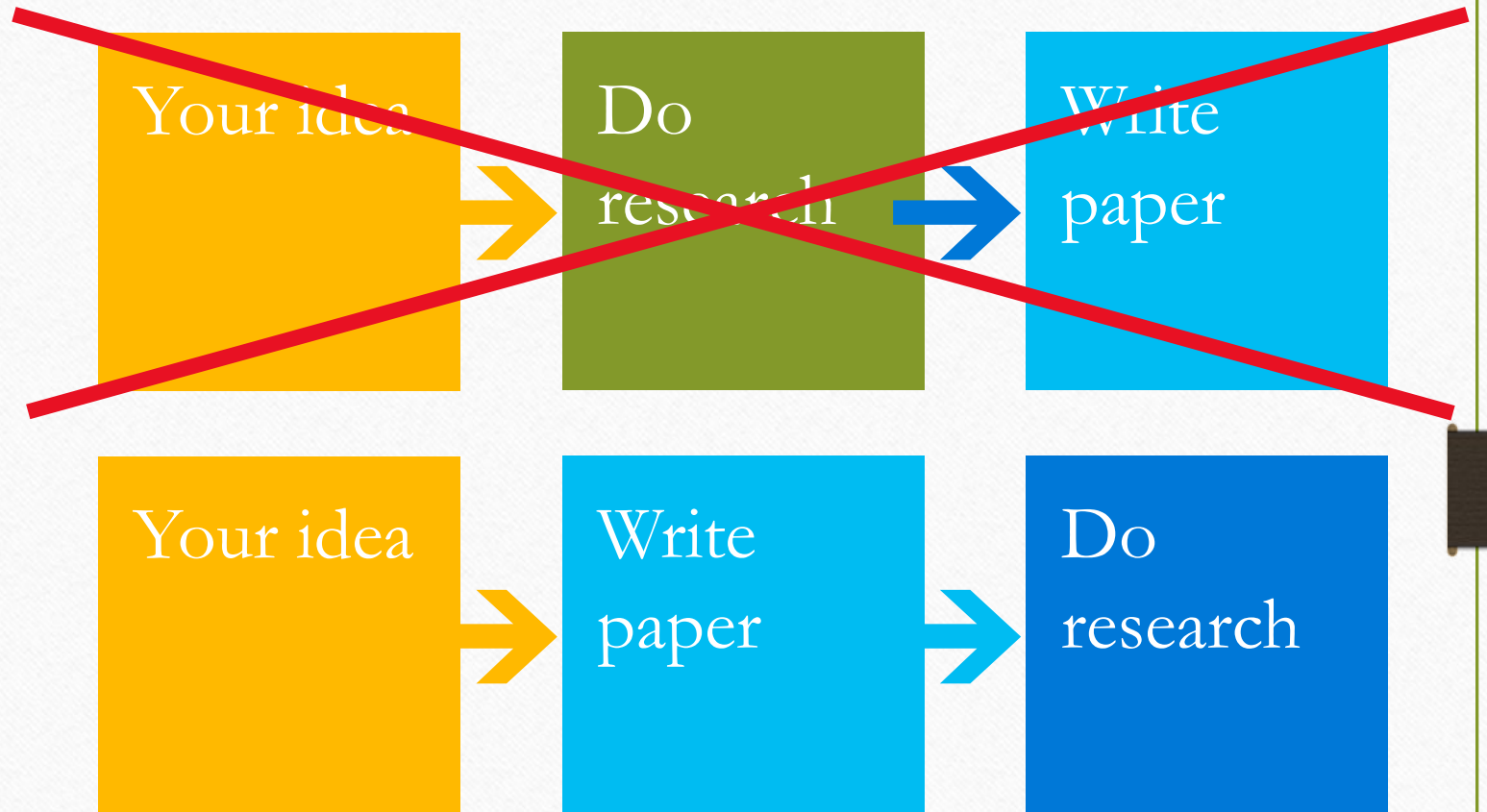


Don't
wait: write

Writing papers: model 1



Writing papers: model 2



Writing papers: model 2



Forces us to be clear, focused

Crystallises what we don't understand

Opens the way to dialogue with others:
reality check, critique, and collaboration

Writing papers: model 2



Writing papers is a primary mechanism for doing research (not just for reporting it)



Identify
your key
idea

Your goal: to
convey a **useful**
and **re-usable** idea

- You want to infect the mind of your reader with **your idea**, like a virus

The greatest ideas are (literally) worthless if you keep them to yourself

Do not be
intimidated

Fallacy

You need to have a fantastic idea before you can write a paper. (Everyone else seems to.)

Write a paper, and give a talk, about **any idea**, no matter how weedy and insignificant it may seem to you

Do not be
intimidated

- Writing the paper is how you develop the idea in the first place
- It usually turns out to be more interesting and challenging than it seemed at first

Write a paper, and give a talk, about **any idea**, no matter how weedy and insignificant it may seem to you

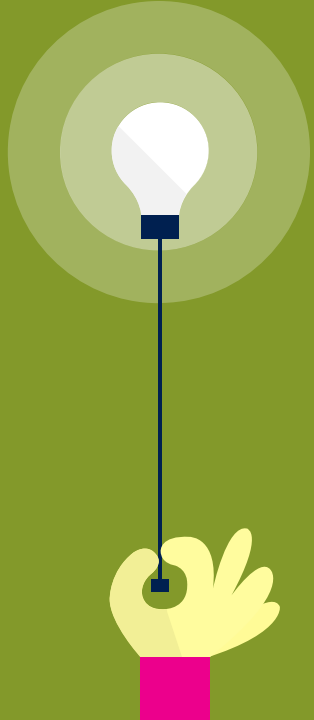
The idea

Idea:

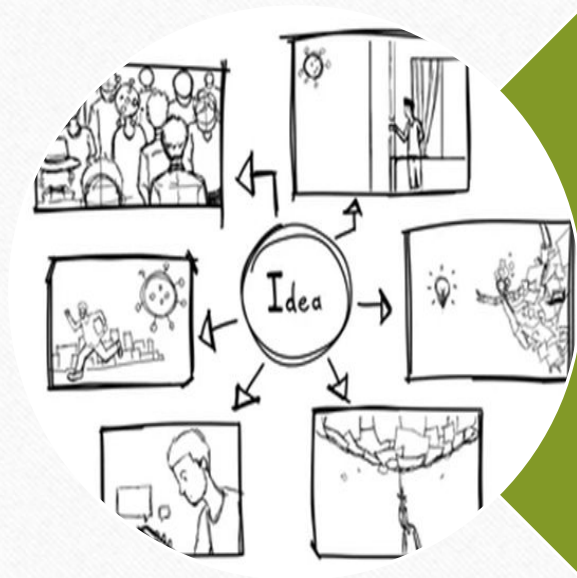
A re-usable insight,
useful to the reader

- Your paper should have just one “ping”: one clear, sharp idea
- You may not know exactly what the ping is when you start writing; but you must know when you finish
- If you have lots of ideas, write lots of papers

Can you hear the “ping”?



- Many papers contain good ideas, but do not distil what they are.
- Make certain that the reader is in no doubt what the idea is. Be 100% explicit:
 - “The main idea of this paper is....”
 - “In this section we present the main contributions of the paper.”



Tell a Story

Your narrative flow

Imagine you are explaining at a whiteboard

- Here is a problem
- It's an interesting problem
- It's an unsolved problem
- Here is my idea
- My idea works (details, data)
- Here's how my idea compares to other people's approaches

Structure (conference paper)

- Title (1000 readers)
- Abstract (4 sentences, 100 readers)
- Introduction (1 page, 100 readers)
- The problem (1 page, 10 readers)
- My idea (2 pages, 10 readers)
- The details (5 pages, 3 readers)
- Related work (1-2 pages, 10 readers)
- Conclusions and further work (0.5 pages)

Contrast with the IMRaD format....



Nail your
contributions
to the mast

The introduction (1 page)

- Describe the problem
- State your contributions

RECALL



...and that is all

5Qs

What is the problem

(CS)

Why is it interesting and important?

ONE PAGE!

Why is it hard?

Why hasn't it been solved before?

What are the key components of my approach and results?

Describe the problem

1 Introduction

There are two basic ways to implement function application in a higher-order language, when the function is unknown: the *push/enter* model or the *eval/apply* model [11]. To illustrate the difference, consider the higher-order function **zipWith**, which zips together two lists, using a function **k** to combine corresponding list elements:

```
zipWith :: (a->b->c) -> [a] -> [b] -> [c]
zipWith k []      []      = []
zipWith k (x:xs) (y:ys) = k x y : zipWith xs ys
```

Here **k** is an *unknown function*, passed as an argument; global flow analysis aside, the compiler does not know what function **k** is bound to. How should the compiler deal with the call **k x y** in the body of **zipWith**? It can't blithely apply **k** to two arguments, because **k** might in reality take just one argument and compute for a while before returning a function that consumes the next argument; or **k** might take three arguments, so that the result of the **zipWith** is a list of functions.

Use an example to introduce the problem

State your contributions

- Write the list of contributions first
- The list of contributions drives the entire paper: the paper substantiates the claims you have made
- Reader thinks “gosh, if they can really deliver this, that’s be exciting; I’d better read on”

State your contributions

Do not leave the reader to guess what your contributions are!

Which of the two is best in practice? The trouble is that the evaluation model has a pervasive effect on the implementation, so it is too much work to implement both and pick the best. Historically, compilers for strict languages (using call-by-value) have tended to use eval/apply, while those for lazy languages (using call-by-need) have often used push/enter, but this is 90% historical accident — either approach will work in both settings. In practice, implementors choose one of the two approaches based on a qualitative assessment of the trade-offs. In this paper we put the choice on a firmer basis:

- We explain precisely what the two models are, in a common notational framework (Section 4). Surprisingly, this has not been done before.
- The choice of evaluation model affects many other design choices in subtle but pervasive ways. We identify and discuss these effects in Sections 5 and 6, and contrast them in Section 7. There are lots of nitty-gritty details here, for which we make no apology — they were far from obvious to us, and articulating these details is one of our main contributions.

In terms of its impact on compiler and run-time system complexity, eval/apply seems decisively superior, principally because push/enter requires a stack like no other: stack-walking

Bulleted list
of
contributions

Contributions should be refutable

No!

We describe the WizWoz system. It is really cool.

We study its properties

We have used WizWoz in practice

Yes!

We give the syntax and semantics of a language that supports concurrent processes (Section 3). Its innovative features are...

We prove that the type system is sound, and that type checking is decidable (Section 4)

We have built a GUI toolkit in WizWoz, and used it to implement a text editor (Section 5). The result is half the length of the Java version.

Evidence



- Your introduction makes claims
- The body of the paper provides evidence to support each claim
- Check each claim in the introduction, identify the evidence, and forward-reference it from the claim
- “Evidence” can be: analysis and comparison, theorems, measurements, case studies

No “rest of
this paper
is...”

- Not:
“The rest of this paper is structured as follows. Section 2 introduces the problem. Section 3 ...Finally, Section 8 concludes”.
- Instead, use forward references from the narrative in the introduction. The introduction (including the contributions) should survey the whole paper, and therefore forward reference every important part.



Related
work: later



Structure

- Abstract (4 sentences)
- Introduction (1 page)
- Related work
- The problem (1 page)
- My idea (2 pages)
- The details (5 pages)
- Conclusions and further work (0.5 pages)

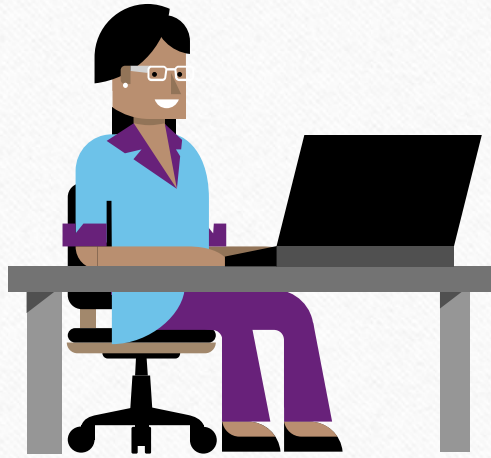


Structure

- Abstract (4 sentences)
- Introduction (1 page)
- The problem (1 page)
- My idea (2 pages)
- The details (5 pages)
- Related work (1-2 pages)
- Conclusions and further work (0.5 pages)

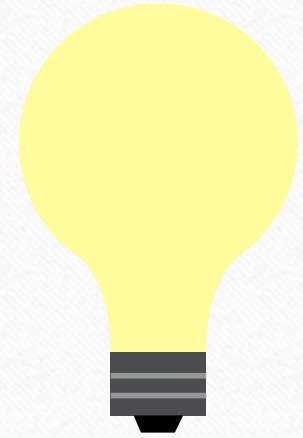


No related
work yet!



Your reader

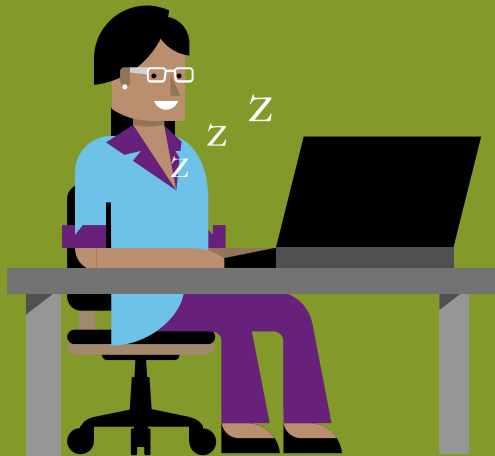
Related work



Your idea

We adopt the notion of transaction from Brown [1], as modified for distributed systems by White [2], using the four-phase interpolation algorithm of Green [3]. Our work differs from White in our advanced revocation protocol, which deals with the case of priority inversion as described by Yellow [4].

No related work yet!



- **Problem 1:** the reader knows nothing about the problem yet; so your (highly compressed) description of various technical tradeoffs is absolutely incomprehensible
- **Problem 2:** describing alternative approaches gets between the reader and your idea

Credit

Fallacy

To make my work look good, I have to make other people's work look bad.

The truth: credit is not like money

- Warmly acknowledge people who have helped you
- Be generous to the competition.
“In his inspiring paper [Foo98] Foogole shows.... We develop his foundation in the following ways...”
- Acknowledge weaknesses in your approach

Giving credit to others does not diminish the credit you get from your paper



Put your
readers
first

Structure

- Abstract (4 sentences)
- Introduction (1 page)
- The problem (1 page)
- My idea (2 pages)
- The details (5 pages)
- Related work (1-2 pages)
- Conclusions and further work (0.5 pages)

Structure

3. The idea

Consider a bifurcated semi-lattice D , over a j i hyper-modulated signature S . Suppose p_i is an element of D . Then we know for every such p_i

- Sounds impressive...but there is an epi-modulus j , such that $p < p$.
- Sends readers to sleep, and/or makes them feel stupid

Presenting the idea

- Explain it as if you were speaking to someone using a whiteboard
- Conveying the intuition is primary, not secondary
- Once your reader has the intuition, she can follow the details (but not vice versa)
- Even if she skips the details, she still takes away something valuable

Conveying the intuition

Introduce the problem, and your idea, using **EXAMPLES** and only then present the general case

- Remember: explain it as if you were speaking to someone using a whiteboard

Using examples

The Simon PJ question: is
there any typewriter font?

2 Background

To set the scene for this paper, we begin with a brief overview of the *Scrap your boilerplate* approach to generic programming. Suppose that we want to write a function that computes the size of an arbitrary data structure. The basic algorithm is “for each node, add the sizes of the children, and add 1 for the node itself”. Here is the entire code for `gsize`:

```
gsize :: Data a => a -> Int
gsize t = 1 + sum (gmapQ gsize t)
```

The type for `gsize` says that it works over any type `a`, provided `a` is a *data* type — that is, that it is an instance of the class `Data`¹. The definition of `gsize` refers to the operation `gmapQ`, which is a method of the `Data` class:

```
class Typeable a => Data a where
  ...other methods of class Data...
  gmapQ :: (forall b. Data b => b -> r) -> a -> [r]
```

Example right
away

Putting the reader first

- Do not recapitulate your personal journey of discovery. This route may be soaked with your blood, but that is not interesting to the reader.
- Instead, choose the most direct route to the idea.



Listen to
your
readers

Getting help

- Experts are good
- Non-experts are also very good
- Each reader can only read your paper for the first time once! So use them carefully
- Explain carefully what you want (“I got lost here” is much more important than “Jarva is mis-spelt”).)

Get your paper read by as many friendly guinea pigs as possible

Getting expert help

- A good plan: when you think you are done, send the draft to the competition saying “could you help me ensure that I describe your work fairly?”.
- Often they will respond with helpful critique (they are interested in the area)
- They are likely to be your referees anyway, so getting their comments or criticism up front is Jolly Good.

Listening to your reviewers

Treat every review like gold dust

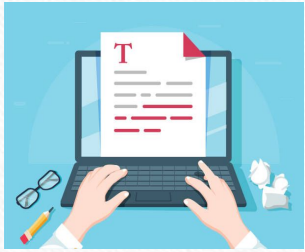
Be (truly) grateful for criticism as well
as praise

This is really, really, really hard

But it's really, really, really, really, really, really,
really, really, really, really important

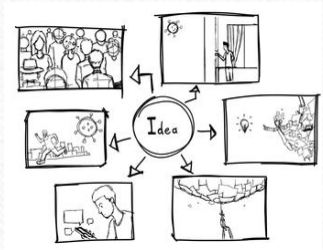
Listening to your reviewers

- Read every criticism as a positive suggestion for something you could explain more clearly
- DO NOT respond “you stupid person, I meant X”.
- INSTEAD: fix the paper so that X is apparent even to the stupidest reader.
- Thank them warmly. They have given up their time for you.



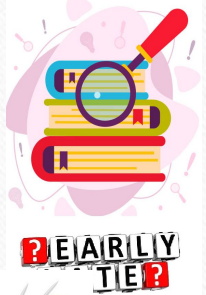
1. Don't wait:
write

2. Identify
your key idea



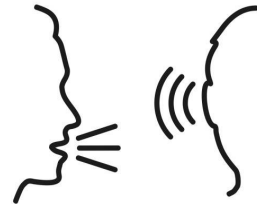
3. Tell a story

4. Nail your
contributions



5. Related
work: later

6. Put your
readers first
(examples)



7. Listen to
your readers

Seven simple,
actionable
suggestions

SUMMARY

Section 3: Language and Style

(not comprehensive)

Basic stuff

- Submit by the deadline
- Keep to the length restrictions
 - Do not narrow the margins
 - Do not use 6pt font
 - On occasion, supply supporting evidence (e.g. experimental data, or a written-out proof) in an appendix
- Always use a spell checker

Visual structure

- Give strong visual structure to your paper using
 - sections and sub-sections
 - bullets
 - italics
 - laid-out code
- Find out how to draw pictures, and use them

Visual structure

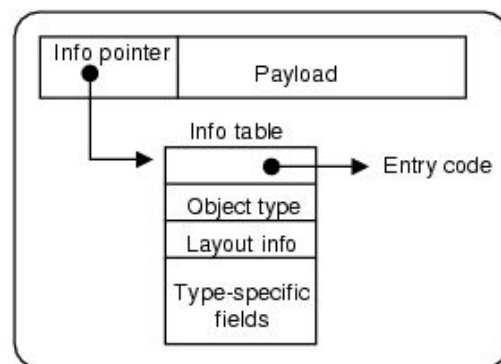


Figure 3. A heap object

The three cases above do not exhaust the possible forms of f . It might also be a *THUNK*, but we have already dealt with that case (rule *THUNK*). It might be a *CON*, in which case there cannot be any pending arguments on the stack, and rules *UPDATE* or *RET* apply.

4.3 The eval/apply model

The last block of Figure 2 shows how the eval/apply model deals with function application. The first three rules all deal with the case of a *FUN* applied to some arguments:

- If there are exactly the right number of arguments, we behave exactly like rule *KNOWNCALL*, by tail-calling the function. Rule *EXACT* is still necessary — and indeed has a direct counterpart in the implementation — because the function might not be statically known.
- If there are too many arguments, rule *CALLK* pushes a *call*

remainder of the object is called the *payload*, and may consist of a mixture of pointers and non-pointers. For example, the object $CON(C\ a_1 \dots a_n)$ would be represented by an object whose info pointer represented the constructor C and whose payload is the arguments $a_1 \dots a_n$.

The info table contains:

- Executable code for the object. For example, a *FUN* object has code for the function body.
- An object-type field, which distinguishes the various kinds of objects (*FUN*, *PAP*, *CON* etc) from each other.
- Layout information for garbage collection purposes, which describes the size and layout of the payload. By “layout” we mean which fields contain pointers and which contain non-pointers, information that is essential for accurate garbage collection.
- Type-specific information, which varies depending on the object type. For example, a *FUN* object contains its arity; a *CON* object contains its constructor tag, a small integer that distinguishes the different constructors of a data type; and so on.

In the case of a *PAP*, the size of the object is not fixed by its info table; instead, its size is stored in the object itself. The layout of its fields (e.g. which are pointers) is described by the (initial segment of) an argument-descriptor field in the info table of the *FUN* object which is always the first field of a *PAP*. The other kinds of heap object all have a size that is statically fixed by their info table.

A very common operation is to jump to the entry code for the object, so GHC uses a slightly-optimised version of the representation in Figure 3. GHC places the info table at the addresses *immediately*

Use the active voice

The passive voice is “respectable” but it **deadens** your paper. Avoid it at all costs.

No!

It can be seen that...

34 tests were run

These properties were thought desirable

It might be thought that this would be a type error

Yes!

We can see that...

We ran 34 tests

We wanted to retain these properties

You might think this would be a type error

Use simple,
direct language

No!

The object under study was
displaced horizontally

On an annual basis

Endeavour to ascertain

It could be considered that
the speed of storage
reclamation left something
to be desired

Yes!

The ball moved sideways

Yearly

Find out

The garbage collector was
really slow

